Safety Data Sheet

Anti Scale for Cooling Tower

1. Identification of the Substance or Mixture and of the Supplier

Identification:

- **Product name:** Anti Scale for Cooling Tower
- **Synonyms:** 1-Hydroxyethylidene-1,1-Diphosphonic Acid (HEDP), Tolyltriazole (TTA)
- **Formula:** -

Special identification: -

Relevant identified uses of the substance or mixture and uses advised against: Laboratory and industrial use

Details of the supplier of the safety data sheet:

- Prime Atosphere (Thailand) Co., Ltd.
- 888/66 Moo 3, Bang Pu Mai, Mueang Samut Prakan, Samut Prakan 10280, Thailand
- Tel:
- Emergency Telephone Number:

2. Hazards Identification

Classification of the substance/mixture:

Classification according to EC regulation 1272/2008 (CLP)

- Met. Corr. 1; H290 May be corrosive to metals
- Eye Irrit. 2; H319 Causes serious eye irritation

Classification according to Directive 67/548/EEC or 1999/45/EC

- Xi; R36: Irritant; Irritating to eyes

Label Elements: Labelling (CLP)



Signal word: Warning

Hazard statements:

- H290 May be corrosive to metals
- H319 Causes serious eye irritation

Precautionary statements:

- P264: Wash skin thoroughly after handling
- P280: Wear protective gloves/protective clothing/eye protection/face protection
- P301 + P330 + P331: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

- P304 + P340: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
- P303 + P361 + P353: IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water or shower
- P305 + P351 + P338: IF IN EYES: Rinse continuously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing
- P403: Store in a well ventilated place
- P405: Store locked up

Other hazards: -

3. Composition/Information on Ingredients

Chemical Name	CAS No.	Weight Percent (%)
1-Hydroxy Ethylidene-1,1-Diphosphonic Acid	2809-21-4	> 30%

4. First Aid Measures

- **After swallowing:** Immediately make victim drink water (two glasses at most). Do not induce vomiting. Consult a doctor.
- **In case of skin contact:** Take off immediately all contaminated clothing. Rinse skin with water/shower.
- **After inhalation:** Fresh air.
- **After eye contact:** Rinse out with plenty of water. Immediately call in ophthalmologist. Remove contact lenses.

5. Fire Fighting Measures

- **Unsuitable extinguishing media:** Water jet
- **Suitable extinguishing media:** Water fog, Foam, Carbon dioxide (CO_2), Dry powder
- **Special hazards arising from the substance or mixture:** -
- **Advice for firefighters:** SCBA

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures: Evacuate the danger area. Remove ignition sources. Avoid inhalation of dusts/vapour. Avoid substance contact. Ensure adequate ventilation. Do not make sparks and flames.

Personal protection: Goggles/mask/gloves/chemical protective clothing

Environmental precautions: Do not let product enter drains.

Methods and materials for containment and cleaning up:

- Suitable absorbent material: sand, soil, chemical absorbent, fabric
- Cover drains. Collect, bind, and pump off spills. Dispose of properly. Clean up affected area.

7. Handling and Storage

Precautions for safe handling:

- Wear PPE in compliance with section 8
- Do not contact substance directly
- Ensure adequate ventilation
- Do not make sparks and flames

Conditions for safe storage:

- Tightly closed
- Good ventilation
- Dry area
- Keep away from heat and sunlight

8. Exposure Controls/Personal Protection

Control parameters:

- TWA: Inhalative 19.4 mg/m^3; Dermal 4.8 mg/kg bw/d (milligrams per kilogram of body weight per day)
- STEL: -

Suitable engineering control: Good ventilation system

Personal protective equipment:

- **Respiratory protection:** Filter mask
- **Eye/face protection:** Goggles
- **Hand protection:** Gloves
- **Skin protection:** Chemical protective clothing

9. Physical and Chemical Properties

- **Appearance:** Liquid
- **Color:** Colourless to pale, transparent
- **Odor:** Perceptible, aromatic
- **Odor threshold:** -
- **pH:** 2-4 at 25 ℃
- **Melting point/freezing point:** -12 ℃
- **Initial boiling point and boiling range:** > 105 ℃
- **Flash point:** Not available
- **Evaporation rate:** -
- **Flammability:** -
- **Upper/lower flammability limits UEL/LEL (%):** -
- **Vapor pressure:** -
- **Vapor density (Air = 1):** -
- **Relative density (Water = 1):** -
- **Water solubility:** Fully miscible

- **Partition coefficient n-Octanol/Water (log Kow):** -
- **Autoignition temperature:** -
- **Decomposition temperature:** -
- **Viscosity:** -

10. Stability and Reactivity

- **Reactivity:** -
- **Chemical stability:** Stable under standard ambient conditions
- **Possibility of hazardous reactions:** -
- **Conditions to avoid:** Flame, spark, heat, moisture, sunlight
- **Incompatible materials:** Strong oxidizing agents
- **Hazardous decomposition products:** See section 5

11. Toxicological Information

Acute Toxicity:

Oral LD50	2910 mg/kg (1-Hydroxy Ethylidene-1,1-Diphosphonic Acid)
Dermal LD50	> 6310 mg/kg (1-Hydroxy Ethylidene-1,1-Diphosphonic Acid)
Inhalation LC50	-

12. Ecological Information

Toxicity:

LC50	Cyprinodon variegatus: 8132 mg/L/96h (OECD 203)
	Ictalurus punctatus: 1212 mg/L/96h (OECD 203)
	Oncorhynchus mykiss: 160 mg/L/96h
	Oncorhynchus mykiss: 23 mg/L/60h
EC50	Acartia tonsa: 94 mg/L/48h
IC50	-

Persistence and degradability:

1. ca. 5%/28 d (not readily biodegradable according to OECD criteria)
2. 22-23%/28 d (OECD 301D). Poorly biodegradable
3. EC0 Bacteria in activated sludge: 200 mg/L/30 min

13. Disposal Considerations

- **Chemical and contaminated container disposal:** Disposal comply with local regulation

14. Transport Information

- **UN number (ADR/RID, IMDG, IATA):** 3265
- **Transport hazard class:** Class 8 for all modes (road, sea, air)
- **Packing group (ADR/RID, IMDG, IATA):** III

15. Regulatory Information

- Hazardous Substance Act B.E. 2535 (1992)
- Factory Act B.E. 2535 (1992)
- Notification of the Department of Industrial Works: Manual for Storage of Chemicals and Hazardous Substances B.E. 2550 (2007)
- Occupational Safety, Health and Environment Act B.E. 2554 (2011)

16. Other Information

- **Revised Date:** 06/04/2026